

Population Mean Estimation for Teacher-Reported Mental Health

Using Ensemble-Based Multiply Robust (EBMR) Estimator

Mental Health Data Analysis

2026-01-21

Contents

1	Notation	2
2	Executive Summary	2
3	Introduction	2
3.1	Background	2
4	Candidate Propensity Score Models	2
4.1	Model Specifications	2
4.2	Model Estimation Results	3
4.2.1	Model 1: use X_1 (Health) and X_2 (Father)	3
4.2.2	Model 2: use X_1 (Health) and X_3 (Parent Report)	3
4.2.3	Model 3: use X_3 (Parent Report) and X_2 (Father)	4
4.3	Model Weights	4
5	Population Mean Estimates	4
5.1	Main Results	4
5.2	All Model Combinations	5
6	Standard Error Comparison	5
7	Sensitivity Analysis	6
7.1	Methodology	6
7.2	Results	7
8	Conclusions	8
8.1	Key Findings	8

1 Notation

Throughout this report, we use the following notation for variables:

Symbol	Variable Name	Description
Y	teacher_report	Outcome variable: teacher-reported mental health score
X_1	health	Child’s health status
X_2	father	Father presence indicator
X_3	parent_report	Parent-reported mental health score
R	response	Response indicator: $R = 1$ if Y is observed, $R = 0$ if missing
X	(X_1, X_2, X_3)	Vector of all covariates

2 Executive Summary

This report presents an analysis of population mean estimation for teacher-reported mental health outcomes using the Ensemble-Based Multiply Robust (EBMR) method. The analysis addresses the challenge of **missing not at random (MNAR)** data, where missingness may depend on the unobserved outcome itself.

Key Findings:

- **Sample size:** 2,486 observations with **42.7% missing** teacher reports
- **Complete case mean:** 0.1832 (SE = 0.0103)
- **EBMR-adjusted mean:** 0.2113 (SE = 0.0369)
- **Adjustment difference:** +0.0282 (15.4% higher than complete case)
- The EBMR method assigns **99.9% weight to Model 1**, indicating this model is relatively correctly specified for the missing data mechanism

3 Introduction

3.1 Background

Teacher-reported mental health assessments are commonly used in educational and clinical settings. However, missing data is pervasive in such surveys, and the missingness mechanism may not be random. If teachers are more (or less) likely to report on students with certain mental health profiles, naive complete-case analysis will produce biased estimates.

4 Candidate Propensity Score Models

4.1 Model Specifications

We specified three MNAR propensity score models, each including the outcome variable Y and different covariates:

The three propensity score models are specified as follows:

Model 1 (π_1): $R \sim Y + X_1 + X_2 + Y \times X_1 + Y \times X_2$

- Includes health status (X_1) and father presence (X_2) with outcome interactions
- Implicitly assume that X_3 (parent_report) is the IV.
- $h_{\alpha_1}(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3)^\top$

Model 2 (π_2): $R \sim Y + X_1 + X_3 + Y \times X_1 + Y \times X_3$

- Includes health status (X_1) and parent report (X_3) with outcome interactions
- Implicitly assume that X_2 (father) is the IV.
- $h_{\alpha_2}(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3)^\top$

Model 3 (π_3): $R \sim Y + X_3 + X_2 + Y \times X_3 + Y \times X_2$

- Includes parent report (X_3) and father presence (X_2) with outcome interactions
- Implicitly assume that X_1 (health) is the IV.
- $h_{\alpha_3}(X) = (1, X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3)^\top$

Ensemble Step: The ensemble weights ν are estimated using:

$$h_\nu(X) = (X_1, X_2, X_3, X_2X_3, X_1X_2, X_1X_3, X_1X_2X_3)^\top,$$

which differs from $h_{\alpha_j}(X)$ by adding the three-order interaction $X_1X_2X_3$.

4.2 Model Estimation Results

4.2.1 Model 1: use X_1 (Health) and X_2 (Father)

Table 2: Model 1 Parameter Estimates (Convergence: 0, Objective: 1.70e-04)

Variable	Estimate	SE	z	p-value	Sig.
(Intercept)	-0.6335	0.2477	-2.558	0.0105	*
Y	1.4005	0.6011	2.330	0.0198	*
Y:X1	-2.4616	1.2180	-2.021	0.0433	*
Y:X2	-0.9906	0.6336	-1.564	0.1179	
X1	0.2966	0.2528	1.173	0.2407	
X2	0.4827	0.1637	2.948	0.0032	**

4.2.2 Model 2: use X_1 (Health) and X_3 (Parent Report)

Table 3: Model 2 Parameter Estimates (Convergence: 0, Objective: 4.03e-03)

Variable	Estimate	SE	z	p-value	Sig.
(Intercept)	-6.9875	0.2748	-25.432	0.0000	***
Y	8.8064	0.2993	29.425	0.0000	***
Y:X1	0.2080	0.2571	0.809	0.4183	
Y:X3	4.1743	0.6682	6.247	0.0000	***
X1	-0.3900	0.2276	-1.714	0.0865	.

X3	-5.3551	0.6884	-7.779	0.0000	***
----	---------	--------	--------	--------	-----

4.2.3 Model 3: use X_3 (Parent Report) and X_2 (Father)

Table 4: Model 3 Parameter Estimates (Convergence: 0, Objective: 2.37e-03)

Variable	Estimate	SE	z	p-value	Sig.
(Intercept)	-0.9201	0.9016	-1.021	0.3075	
Y	1.9797	1.7691	1.119	0.2631	
Y:X3	-14.7839	2.0452	-7.229	0.0000	***
Y:X2	-0.1911	0.5269	-0.363	0.7169	
X3	0.9692	0.9106	1.064	0.2872	
X2	0.2985	0.2405	1.241	0.2145	

4.3 Model Weights

The EBMR method assigns data-adaptive weights to each model:

Table 5: EBMR Model Weights

Model	Weight	Percentage
Model 1	0.9988	99.88%
Model 2	0.0000	0.00%
Model 3	0.0012	0.12%

Interpretation: Model 1 receives nearly all the weight (99.88%), suggesting it best captures the missing data mechanism. This indicates that the combination of X_1 (health) and X_2 (father), along with their interactions with the outcome Y , provides the most informative model for the propensity scores.

5 Population Mean Estimates

5.1 Main Results

Table 6: Population Mean Estimates: Complete Case vs. EBMR

Estimator	Estimate	SE	95% CI Lower	95% CI Upper
Complete Case	0.1832	0.0103	0.1631	0.2032
EBMR (use 3 models)	0.2113	0.0369	0.1390	0.2836

Key Finding: The EBMR estimate (0.2113) is **15.4% higher** than the complete case estimate (0.1832), with a difference of 0.0282. This suggests that the complete case analysis underestimates the population mean, possibly because individuals with $Y = 1$ are less likely to have complete data.

5.2 All Model Combinations

Table 7: Population Mean Estimates for All Model Combinations

Label	Models	Estimate	Analytical SE	Bootstrap SE	Perturbation SE
CC	Complete Case	0.1832	0.0103	0.0099	0.0112
100	pi1 only	0.2114	0.0369	0.0405	0.0377
010	pi2 only	0.5299	0.0104	0.1397	0.1442
001	pi3 only	0.2855	0.1530	0.1265	0.1310
110	pi1 + pi2	0.2114	0.0369	0.0408	0.0407
101	pi1 + pi3	0.2113	0.0369	0.0405	0.0379
011	pi2 + pi3	0.5269	0.0164	0.1390	0.1427
111	pi1 + pi2 + pi3	0.2113	0.0369	0.0435	0.0431

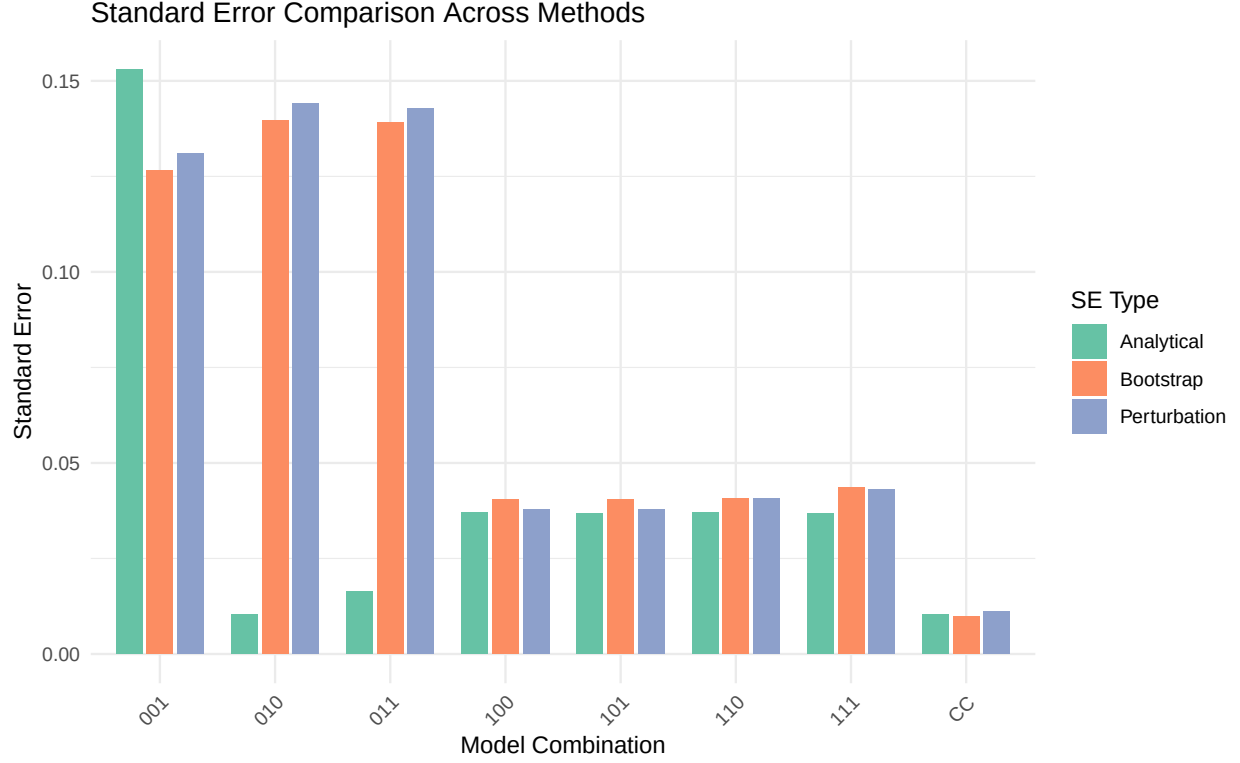
Observations:

1. **Model 1 alone** (label “100”) gives estimate 0.2114, nearly identical to the full ensemble
2. **Model 2 alone** (label “010”) gives a much higher estimate (0.5299) with large bootstrap SE (0.1397), indicating instability
3. **Model 3 alone** (label “001”) gives estimate 0.2855 with the largest analytical SE (0.1530)
4. **Combinations including Model 1** (110, 101, 111) all give similar estimates around 0.2113
5. **Combinations without Model 1** (011) give much higher estimates similar to Model 2

6 Standard Error Comparison

Table 8: Standard Error Comparison

Method	Analytical SE	Bootstrap SE	Perturbation SE
Complete Case	0.0103	0.0099	0.0112
EBMR (111)	0.0369	0.0435	0.0431



Observations:

- For the complete case and Model 1 combinations, all three SE methods give similar results
- – For Model 2 (010) and combinations with Model 2 but without Model 1, bootstrap and perturbation SEs are much larger than analytical SEs, indicating potential model instability

7 Sensitivity Analysis

7.1 Methodology

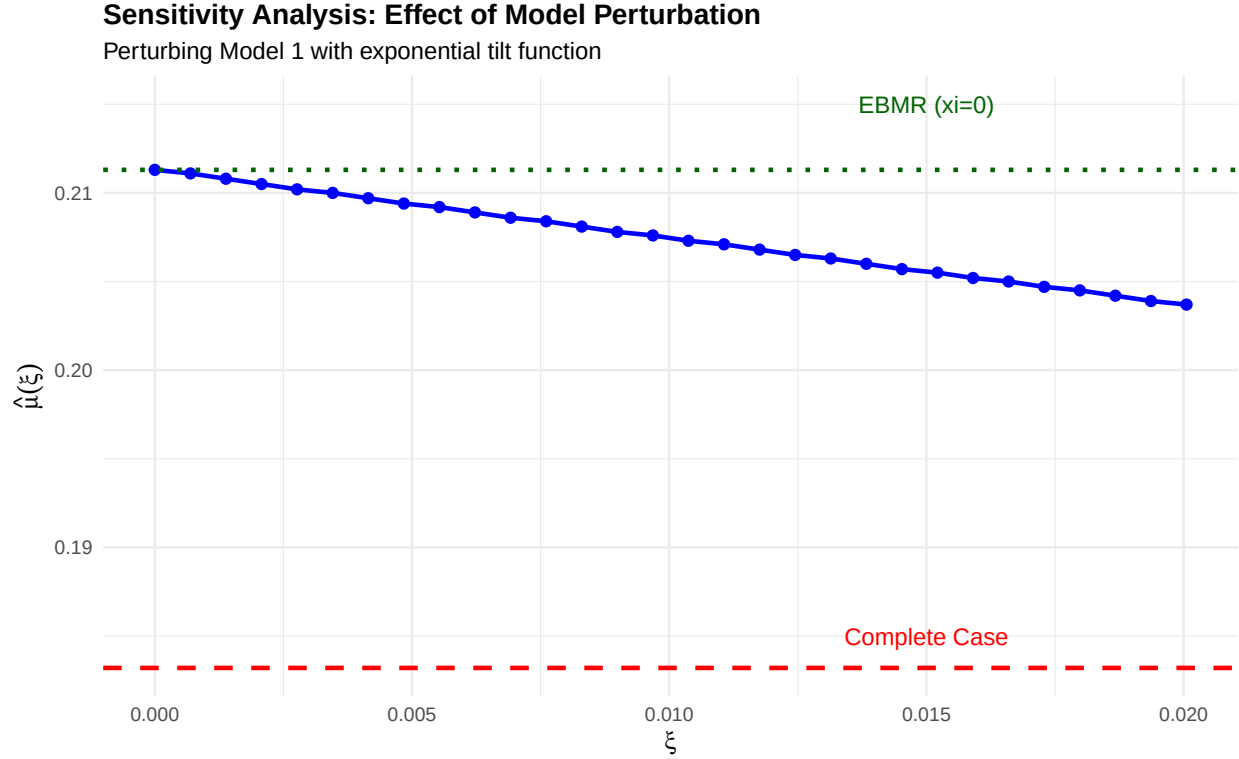
We conducted a sensitivity analysis by perturbing Model 1 using an exponential tilt function:

$$\tilde{\pi}_1(X, Y) = \pi_1(X, Y) \exp(\xi \cdot (Y + X_1 + X_2 + X_3))$$

where ξ ranges from 0 to $n^{-1/2} \approx 0.020$. The resulting estimator is denoted as $\hat{\mu}(\xi)$.

This tests the robustness of our estimates to local misspecification of the propensity score model.

7.2 Results



The sensitivity analysis results are summarized below:

Metric	Value
ξ range	[0, 0.0201]
$\hat{\mu}(\xi)$ range	[0.2037, 0.2113]
$\hat{\mu}(\xi)$ at Baseline ($\xi = 0$)	0.2113
$\hat{\mu}(\xi)$ at max ξ	0.2037
Change from baseline	-0.0076 (-3.6%)

Interpretation:

- As ξ increases from 0 to 0.02, the estimated population mean decreases monotonically from 0.2113 to 0.2037
- The total change is only **-0.0076 (3.6%)**, indicating the estimate is relatively robust to local misspecification
- Even at maximum perturbation, the EBMR estimate (0.2037) remains substantially higher than the complete case estimate (0.1832)
- This provides confidence that the adjustment for MNAR missingness is genuine and not an artifact of model specification

8 Conclusions

8.1 Key Findings

1. **Evidence of MNAR:** The association between outcome and missingness, combined with the difference between complete case and EBMR estimates, suggests the data are MNAR.
2. **Population Mean Adjustment:** The EBMR method estimates the population mean at **0.2113**, which is **15.4% higher** than the complete case estimate of 0.1832. This indicates that complete case analysis underestimates mental health concerns in this population.
3. **Model Ensemble:** The EBMR assigns nearly all weight (99.9%) to Model 1, which includes X_1 (health) and X_2 (father) and implicitly assumes that X_3 (parent_report) is the valid IV.
4. **Robustness:** Sensitivity analysis shows the estimate is robust to local model misspecification, with only a 3.6% change across the perturbation range.

Report generated on 2026-01-21